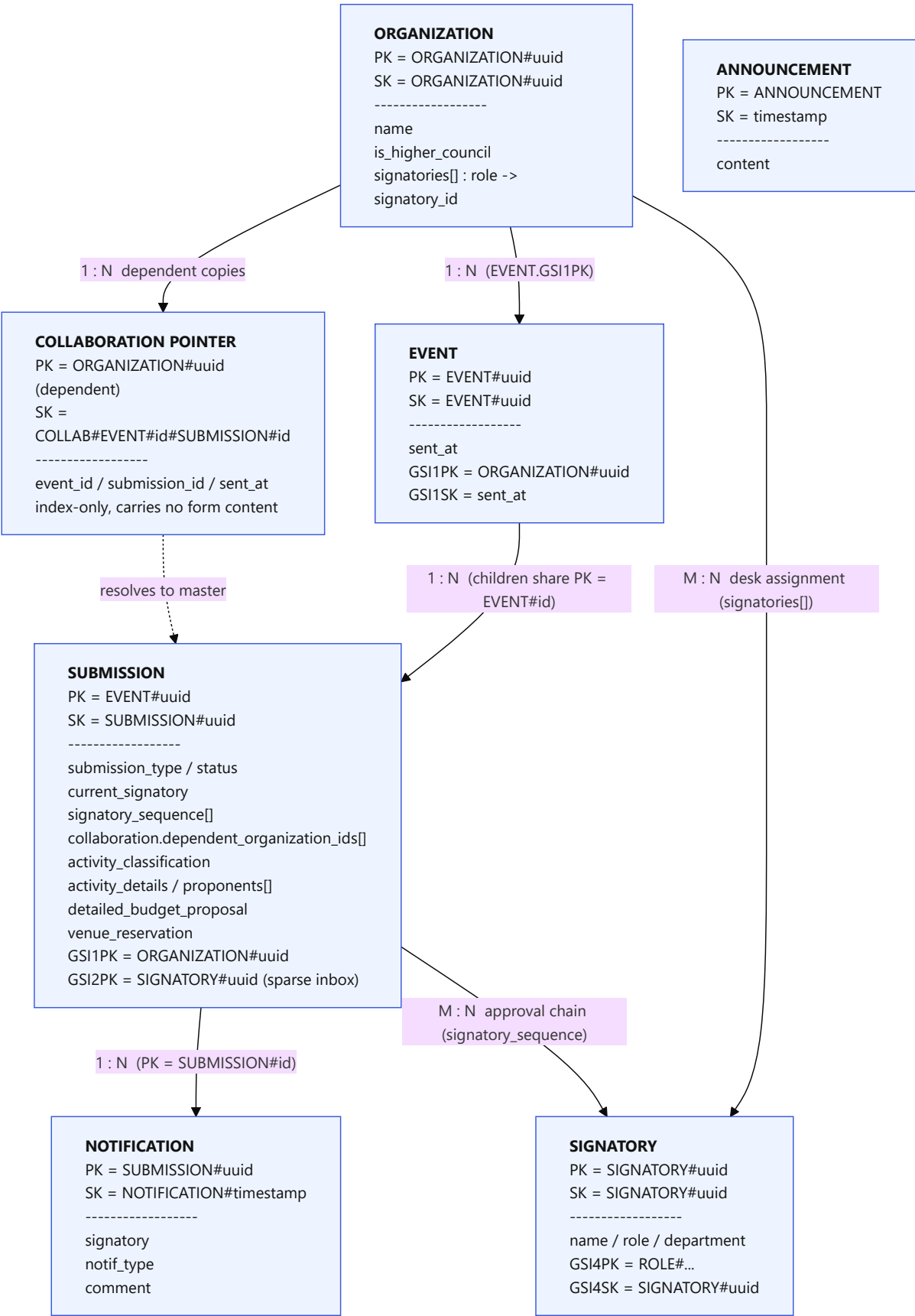


mapua-apex Data Model

DynamoDB single-table data model — conceptual ERD, object dictionary, and access patterns.

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Conceptual ERD



Data Objects & Properties

All objects live in a **single DynamoDB table**. Each item is identified by a partition key (`PK`) and a sort key (`SK`); several item types share one partition so related records are read together in a single query. There are **no foreign keys and no cascade**: related IDs are *copied* onto items and resolved at write time, so deleting one item never rewrites another.

ORGANIZATION

The student organization. Also the desk directory: it records which signatory person occupies each approval role for this org.

Property	Type	Notes
<code>PK</code>	string	<code>ORGANIZATION#uuid</code>
<code>SK</code>	string	<code>ORGANIZATION#uuid</code> (same as PK)
<code>name</code>	string	Display name
<code>is_higher_council</code>	boolean	Default <code>false</code> . When true, the Dean step is skipped in routing
<code>signatories</code>	list	Desk directory, one person per role: <code>{ role, signatory_id }</code>
<code>signatories[].role</code>	string	One of <code>adviser</code> / <code>dean</code> / <code>osaar</code> / <code>cdm</code> / <code>admin</code>
<code>signatories[].signatory_id</code>	string	Plain copy of a SIGNATORY id

Allowed desk roles are stored in the order `adviser` → `dean` → `osaar` → `cdm` → `admin` regardless of write order. `admin` may hold a desk but is **not** part of the approval sequence. `osaar` and `cdm` desks are campus-wide (from `OSAAR_SIGNATORY_ID` / `CDM_SIGNATORY_ID` in the API env), not stored per org.

EVENT

A filing/event that owns one or more submissions. Its partition also holds the event's SUBMISSION children.

Property	Type	Notes
<code>PK</code>	string	<code>EVENT#uuid</code>
<code>SK</code>	string	<code>EVENT#uuid</code> (same as PK)
<code>sent_at</code>	timestamp	Creation time (UTC)
<code>GSI1PK</code>	string	<code>ORGANIZATION#uuid</code> — owning org (also identifies the <i>proponent</i> org for collaborations)

Property	Type	Notes
GSI1SK	timestamp	<code>sent_at</code> — for newest-first org event lists

SUBMISSION

A proposal paper (e.g. SAAF) filed under an event. Carries the full form body plus routing state and index keys. This is the single source of truth for a submission, including any collaboration.

Identity, routing & indexes

Property	Type	Notes
PK	string	<code>EVENT#uuid</code> (parent event partition)
SK	string	<code>SUBMISSION#uuid</code>
<code>submission_type</code>	string	e.g. <code>saaf</code>
<code>sent_at</code>	timestamp	Submit time (UTC)
<code>status</code>	string	<code>pending</code> / <code>approved</code> / <code>denied</code> / <code>returned</code>
<code>current_signatory</code>	string	<code>SIGNATORY#uuid</code> holding the paper now
<code>signatory_sequence</code>	list	Ordered <code>SIGNATORY#uuid</code> routing snapshot, written at submit time
<code>collaboration.dependent_organization_ids</code>	list	<code>ORGANIZATION#uuid</code> dependents; empty/absent = no collaboration
GSI1PK	string	<code>ORGANIZATION#uuid</code> (proponent org)
GSI1SK	string	<code>SUBMISSION#uuid</code>
GSI2PK	string	<code>SIGNATORY#uuid</code> — sparse signatory inbox (present while <code>pending</code> / <code>returned</code>)
GSI2SK	timestamp	Desk queue time

activity_classification

Property	Type	Notes
<code>activity_type</code>	string	<code>co-curricular</code> / <code>extra-curricular</code>
<code>total_org_members</code>	number	

Property	Type	Notes
nature	string	major / minor — set by OSAAR via the classification PATCH, not by the student create body

proponents[] (list of people)

id, position_title, first_name, middle_name, last_name, suffix, student_number, program_and_year, date_of_submission, department, position_of_applicant, org_or_course_section, contact_number, email_address, facebook_link.

activity_details

title_and_nature, description, objectives, venue, date_of_event, day_of_event, time_of_event, expected_participants, individual_contribution, proposed_budget. time_of_event is 24-hour "HH:MM – HH:MM"; saved times fall between 7:00 AM and 9:00 PM, end not earlier than start, and start ≠ end.

institutional_alignment

mission_statements { competitive, research, solutions }, core_values_explanation, peo_explanation, sdg_explanation.

detailed_budget_proposal

items[] { item_no, unit, quantity, price_per_unit, total }, grand_total.

venue_reservation

Property	Type	Notes
has_reservation	boolean	Drives whether the CDM desk joins the routing chain
equipment_requested	object	monoblock_chairs, whiteboards, tables, rostrum, flags_with_stand, panel_boards, others_specified
general_facilities	object	{ purpose, items[] { item, date_of_use, time_of_use, location } }
function_rooms	object	{ purpose, items[] { date_needed, time_needed, room_needed, remarks } }
audiovisual_equipment	object	{ purpose, items[] { date_needed, time_needed, equipment_needed, remarks } }

NOTIFICATION

One row in a submission's approval/timeline history. Stored in a partition keyed by the submission id (a separate partition from the event).

Property	Type	Notes
PK	string	SUBMISSION#uuid
SK	string	NOTIFICATION#timestamp (identity; UTC second precision)
signatory	string	SIGNATORY#uuid author (snapshot)
notif_type	string	approved / fully approved / denied / returned
comment	string	Mandatory when denied or returned

Approve / return / deny insert a NOTIFICATION row as a side effect. A PUT replaces `notif_type` / `comment` in place and does **not** change the SK.

SIGNATORY

A person (the signatory identity). Has **no** `organization_id`; desk assignment lives on `ORGANIZATION.signatories`.

Property	Type	Notes
PK	string	SIGNATORY#uuid
SK	string	SIGNATORY#uuid (same as PK)
name	string	
role	string	adviser / cdm / dean / osaar / admin
department	string	Deans only (e.g. SOIT)
GSI4PK	string	ROLE#ADVISER / ROLE#CDM / ROLE#DEAN / ROLE#DEAN#{ DEPARTMENT} / ROLE#OSAAR / ROLE#ADMIN
GSI4SK	string	SIGNATORY#uuid

Other items copy the signatory id rather than join back to SIGNATORY:

`ORGANIZATION.signatories[].signatory_id`, `SUBMISSION.current_signatory` / `GSI2PK`, and `NOTIFICATION.signatory`. Renaming keeps the same uuid; deleting a SIGNATORY does not rewrite those copies.

COLLABORATION POINTER

An index-only item, one per dependent organization, that makes the proponent's submission discoverable in a dependent's dashboard. It carries **no** form content — the master SUBMISSION stays the single source of truth.

Property	Type	Notes
PK	string	ORGANIZATION#uuid — the dependent org (not the proponent)
SK	string	COLLAB#EVENT#uuid#SUBMISSION#uuid
event_id	string	EVENT#uuid
submission_id	string	SUBMISSION#uuid
sent_at	timestamp	

Queried from the dependent's partition (SK begins COLLAB#); no GSI is needed because DynamoDB cannot index the N values of an array. A dependent may read the shared submission and its NOTIFICATION timeline but cannot edit or resubmit — only the proponent may.

ANNOUNCEMENT

A campus-wide announcement. All announcements share one partition; the SK (timestamp) is the identity. Admin Cognito group only.

Property	Type	Notes
PK	string	ANNOUNCEMENT (constant)
SK	string	timestamp — identity (sent_at in the API); newest-first list
content	string	

A PUT replaces content in place and does not change the SK.

mapua-apex — DynamoDB Access Patterns

DynamoDB is modeled around **access patterns**, not normalized relations: every query is designed first, then the partition key (PK), sort key (SK), and global secondary indexes (GSI) are chosen to serve it. All patterns hit a **single table**. There are no joins — related IDs are copied onto items at write time.

Index summary:

Index	Key	On	Purpose
Base table	PK / SK	all items	Primary access
GSI1	GSI1PK / GSI1SK	EVENT, SUBMISSION	Org-scoped event list & admin submission list
GSI2	GSI2PK / GSI2SK	SUBMISSION	Sparse signatory inbox (pending / returned)
GSI4	GSI4PK / GSI4SK	SIGNATORY	Role directory (ROLE# ...)

Core access patterns

#	Access pattern	Operation	Key condition	Index
1	Org event list (student dashboard)	Query	GSI1PK = ORGANIZATION#id , GSI1SK desc (first 25 EVENT items)	GSI1
2	Event children (its submissions)	Query	PK = EVENT#id , SK begins SUBMISSION#	Base
3	Student's own submissions	Query	GSI1 for org, then each event partition	GSI1 + Base
4	Collaboration submissions (dependent view)	Query	PK = ORGANIZATION# <dep> , SK begins COLLAB# , then resolve each pointer to its master SUBMISSION	Base
5	Single submission detail	GetItem	PK = EVENT#id , SK = SUBMISSION#id	Base
6	Submission timeline / audit	Query	PK = SUBMISSION#id , SK begins NOTIFICATION#	Base

#	Access pattern	Operation	Key condition	Index
7	Signatory inbox (desk queue)	Query	GS12PK = SIGNATORY#id , GS12SK by time (sparse)	GS12
8	Signatory directory by role	Query	GS14PK = ROLE# ... , GS14SK begins SIGNATORY#	GS14
9	Current signatory profile (/signatories/me)	GetItem	PK = SIGNATORY#id , SK = SIGNATORY#id	Base
10	Admin: all submissions	Query	per org GS11PK = ORGANIZATION#id , GS11SK begins SUBMISSION#	GS11
11	Student org record (/students/organization)	GetItem	PK = ORGANIZATION#id , SK = ORGANIZATION#id	Base
12	Org directory (/students/organizations)	Query	org list → { organization_id, name }	Base
13	Announcement list	Query	PK = ANNOUNCEMENT , SK desc (newest first)	Base
14	Announcement show / edit / delete	GetItem / PutItem / DeleteItem	PK = ANNOUNCEMENT , SK = timestamp	Base
15	Write submission (+ collab pointers)	PutItem	PK = EVENT#id , SK = SUBMISSION#id ; maintain one COLLAB pointer per dependent	Base
16	Approve / return / deny	UpdateItem	advance current_signatory / GS12PK from stored signatory_sequence ; write NOTIFICATION	Base

Page / feature → objects accessed

Page / feature	Objects read or written	Patterns
Student dashboard	ORGANIZATION, EVENT, SUBMISSION, ANNOUNCEMENT, COLLAB POINTER	1, 2, 3, 4, 11, 13
Submission tracker / detail	SUBMISSION, NOTIFICATION, SIGNATORY (by copied id)	5, 6
New / edit submission (SAAF)	SUBMISSION, COLLAB POINTER, ORGANIZATION (directory)	12, 15
Signatory desk (inbox)	SUBMISSION, SIGNATORY	7, 9
Signatory review (approve/return/deny/classify)	SUBMISSION, NOTIFICATION	6, 16
Admin dashboard (submissions)	ORGANIZATION, SUBMISSION	10
Admin → Organizations	ORGANIZATION, SIGNATORY	11, 12
Admin → Signatories	SIGNATORY	8
About / Announcements	ANNOUNCEMENT	13, 14